

Log Transformation for Outlier Handling

Log transformation converts the original values into their logarithmic values:

$$X_{\text{new}} = \log(X)$$

The important point is:

Log transformation does not remove the outlier. It compresses the scale of large values, thereby reducing their influence.

For example:

None

Original:

10

20

30

40

50

1000

Using $\log_{10}$:

None
1.00
1.30
1.48
1.60
1.70
3.00

The 1000 is still present, but its effect is significantly compressed.

Its scale and influence are reduced.

When Is Log Transformation Particularly Useful?

It works best when the numerical variable has a **right-skewed distribution**.

Common examples:

- Income Salary
- House prices
- Revenue Sales
- Population
- Transaction amount

- Website traffic
- Number of followers
- Company valuation

`np.log()` vs `np.log10()`

There are different logarithmic bases.

Natural log

None

```
np.log(df["revenue"])
```

Uses base **e**.

Base-10 log

None

```
np.log10(df["revenue"])
```

Uses base **10**.

For outlier handling, the important concept isn't the specific base. The important idea is **compression of large values**.

What About Zero?

A standard logarithm cannot handle zero:

None

```
np.log(0)
```

Result:

None

```
-inf
```

If your data contains zero, you can use:

None

```
np.log1p(df["revenue"])
```

$\log_1p(x)$ calculates:

$\log(1+x)$

So:

None

```
np.log1p(0)
```

gives:

None

0

This makes `log1p()` useful for variables containing **zero values**.

What About Negative Values?

Standard log transformation cannot be directly applied to negative values.

For example:

None

`np.log(-10)`

is not valid in real-valued data.

If the feature contains negative values, consider transformations such as **Yeo-Johnson**, which can handle both positive and negative values.

Important: Don't Apply Log Transformation Blindly

Suppose you identify an outlier using IQR:

None

Income = ₹10 crore

Before transforming, ask:

Is this actually an error?

If it's a genuine observation, deleting it may be inappropriate.

If the variable is strongly right-skewed, log transformation could be a better choice.

So the workflow becomes:

None

Detect outliers



Understand the outlier



Is it an error?

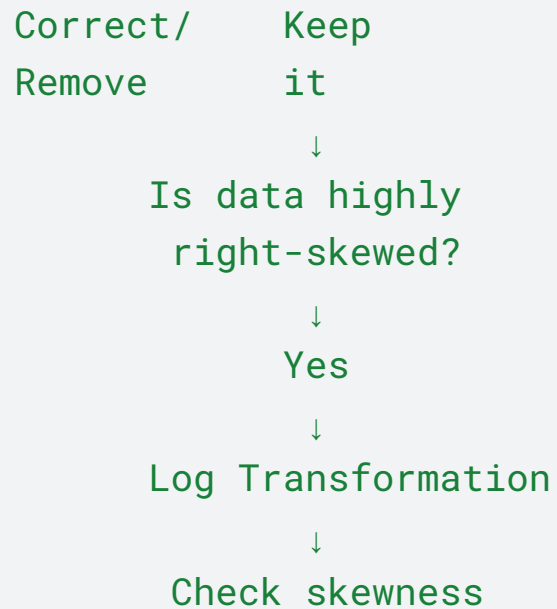


Yes



No





Key takeaway

Z-score and IQR primarily help us identify outliers. Log transformation is a way to transform the data so that extreme large values have less influence.

One final point to emphasize to learners:

Don't say "log transformation removes outliers."

The correct statement is:

"Log transformation reduces the impact of extreme values by compressing the scale, especially for right-skewed data."

